Cranial Nerves (V, VII, IX, X, XII): Set 3

1. A 34-year-old patient undergoes an extraoral surgical approach to repair a submandibular gland pathology. Postoperatively, the patient presents with an inability to depress the corner of the mouth on the affected side, causing a noticeable asymmetry when smiling. This diagnostic sign confirms iatrogenic injury to which specific branch of the facial nerve?

a. Marginal mandibular branch
b. Cervical branch
c. Buccal branch
d. Temporal branch
e. Zygomatic branch

2. A patient experiences a severe basilar skull fracture that completely shears the facial nerve (CN VII) within the internal acoustic meatus. Alongside complete unilateral facial paralysis, which additional clinical manifestation will this intra-cranial checkpoint lesion produce?

a. Hyperacusis, loss of taste on the anterior two-thirds of the tongue, and loss of lacrimation
b. Loss of taste on the posterior one-third of the tongue and parotid dryness
c. Severe numbness over the chin skin and lower lip mucosa
d. Deviation of the tongue toward the injured side
e. Loss of the efferent limb of the pupillary light reflex

3. If a patient presents with an isolated lesion of the facial nerve inside the facial canal immediately *distal* to the geniculate ganglion but *proximal* to the takeoff of the nerve to the stapedius, what functions are preserved?

b. Taste sensation on the anterior tongue
b. Normal lacrimation (tearing) via the greater petrosal nerve
c. Motor control of the orbicularis oculi
d. Sensory perception of the cornea touch
e. Secretomotor drive to the submandibular gland

4. A 29-year-old patient presents with standard Bell's palsy. On clinical evaluation, tears are seen continuously overflowing down the cheek rather than draining normally into the lacrimal puncta (epiphora). This occurs because of paralysis of which specific muscle mechanism?

a. Levator palpebrae superioris muscle sheet
b. Superior tarsal muscle smooth fibers
c. Frontal belly of the occipitofrontalis
d. Palpebral part of the orbicularis oculi muscle, leading to sagging of the lower eyelid
e. Corrugator supercilii fiber loops

5. A patient presents with hyperacusis (normal sounds perceived as unbearably loud). This specific sign isolates a lesion of the facial nerve proximal to its branching within the middle ear cavity to which muscle?

a. Tensor tympani muscle
b. Tensor veli palatini
c. Tensor tympani and stapedius simultaneously
d. Stapedius muscle
e. Tensor veli palatini and palatoglossus

6. A dental surgeon performs an inferior alveolar nerve block to extract a lower impacted third molar. If the needle is accidentally directed too far posteriorly into the

parenchyma of the parotid gland, the anesthetic will block the terminal branches of CN VII, resulting in a transient:
a. Ipsilateral facial paralysis (inability to close the eye or wrinkle the forehead on that side)
b. Total loss of taste on the anterior tongue
c. Profuse salivation from Stensen's duct
d. Severe, stabbing pain radiating up to the temporal fossa
e. Devastating numbness over the mandibular teeth
7. A 52-year-old patient presents with severe, sharp paroxysms of stabbing, electric shock-like pain triggered by eating, washing the face, or shaving. The pain radiates along the mandibular body skin. This diagnostic condition is:
a. Bell's palsy
b. Trigeminal neuralgia (Tic douloureux affecting V3)
c. Glossopharyngeal neuralgia
d. Frey's syndrome
e. Horner's syndrome
8. To relieve intractable trigeminal neuralgia affecting the mandibular division, a neurosurgeon decides to selectively ablate the nerve fibers. The surgical needle can access V3 extraorally as it emerges from the skull base by targeting which opening?
a. Foramen rotundum
b. Foramen spinosum
c. Superior orbital fissure
d. Foramen ovale
e. Jugular foramen
9. A patient presents with a deep tumor infiltrating the infratemporal fossa roof. Clinical examination reveals total loss of general somatic sensation over the anterior two-thirds of the tongue, the mucous membrane of the floor of the mouth, and all mandibular teeth on that side. This isolates a lesion of the:
a. Maxillary nerve (V2)
b. Lingual nerve proximal to its junction with the chorda tympani
c. Mandibular nerve (V3) main trunk or sensory divisions
d. Glossopharyngeal nerve (CN IX)
e. Hypoglossal nerve (CN XII)
10. In the same patient with a complete V3 lesion, when they are asked to open their mouth wide, the mandible deviates characteristically to the paralyzed side. This mechanical deviation is driven by the unopposed action of the healthy contralateral:
a. Masseter muscle
b. Temporalis posterior fibers
c. Medial pterygoid muscle sling
d. Lateral pterygoid muscle
e. Anterior belly of digastric
11. A 42-year-old patient presents with a severe malignant tumor of the palatine tonsil. The patient experiences agonizing pain radiating deeply into the middle ear cavity when swallowing. This referred pain pathway tracks through which nerve?
a. Trigeminal nerve (V2 division)
b. Facial nerve (CN VII via chorda tympani)
c. Vagus nerve (CN X)
d. Glossopharyngeal nerve (CN IX via the tympanic branch/Jacobson's nerve)
e. Hypoglossal nerve (CN XII)

12. A physician tests a patient's gag reflex by touching the posterior wall of the oropharynx with a tongue depressor. The patient exhibits no sensory awareness and no muscular contraction. This complete loss of the reflex indicates a combined lesion of which two cranial nerves?
a. CN V3 and CN VII
b. CN VII and CN IX
c. CN IX (afferent limb) and CN X (efferent limb)
d. CN X and CN XII
e. CN V2 and CN XII
13. During a difficult surgical excision of an acoustic neuroma at the cerebellopontine angle, a surgeon accidentally cuts a nerve that leads to an absolute loss of the cough reflex superior to the vocal folds, a deviated uvula pointing to the healthy side, and a drooping soft palate. This nerve is the:
a. Glossopharyngeal nerve (CN IX)
b. Vagus nerve (CN X)
c. Spinal accessory nerve (CN XI)
d. Hypoglossal nerve (CN XII)
e. Facial nerve (CN VII)
14. A patient presents with hoarseness of the voice following a surgical procedure for a thyroid goiter. Direct laryngoscopy confirms that the right vocal fold is fixed in the paramedian position due to a lesion of the:
a. Superior laryngeal nerve internal branch
b. External laryngeal nerve supplying cricothyroid
c. Recurrent laryngeal nerve (branch of CN X)
d. Pharyngeal branch of vagus
e. Hypoglossal nerve trunk
15. A patient presents with a deep stab wound in the upper neck. On clinical examination, when asked to protrude their tongue, the tongue tip deviates sharply to the right side. Atrophic wasting and fasciculations are noted on the right half of the tongue body. This isolates a lower motor neuron lesion of the:
a. Left hypoglossal nerve
b. Left lingual nerve
c. Right hypoglossal nerve (CN XII)
d. Right glossopharyngeal nerve
e. Left vagus nerve trunk
16. Let's briefly touch on other nerves. A patient presents with complete ptosis (drooping upper eyelid), a fully dilated pupil that does not react to light, and an eyeball that rests in a "down-and-out" position. This presentation matches a lesion of:
a. CN IV (Trochlear)
b. CN VI (Abducens)
c. CN III (Oculomotor nerve)
d. The sympathetic trunk superior ganglion
e. CN V1 (Ophthalmic)
17. A patient presents with a complete inability to look downward and inward (toward the nose), making it highly difficult to walk down stairs or read a book. This specific isolated physical deficit isolates a lesion of:
a. CN III (Oculomotor)
b. CN IV (Trochlear nerve supplying the superior oblique muscle)

c. CN VI (Abducens)
d. CN V1 (Nasociliary branch)
e. CN VII (Temporal branch)

18. A 60-year-old patient presents with a severe internal strabismus (medial deviation) of the left eye. When asked to look left, the left eye cannot abduct past the midline. This isolates a lesion of:

a. Left oculomotor nerve
b. Left trochlear nerve
c. Left abducens nerve (CN VI supplying the lateral rectus muscle)
d. Left ophthalmic nerve
e. Left facial nerve

19. A patient presents with unilateral sensory loss over the forehead, bridge of the nose, and upper eyelid, along with a complete loss of the afferent path of the corneal reflex. This indicates a selective lesion of the:

a. Maxillary nerve (V2)
b. Ophthalmic nerve (V1 division of the Trigeminal nerve)
c. Mandibular nerve (V3)
d. Facial nerve motor root
e. Optic nerve tract

20. A patient presents with a profound loss of hearing and unresolving vertigo following a temporal bone fracture. The trauma most likely damaged which cranial nerve within the petrous canal?

a. CN VII
b. CN VIII (Vestibulocochlear nerve)
c. CN IX
d. CN X
e. CN XI

21. A patient presents with a rare condition called Frey's syndrome (lucid gustatory sweating) following a traumatic surgical removal of a parotid gland tumor. When the patient eats, the skin over the temple beads with sweat and turns red because regenerated parasympathetic secretomotor fibers mistakenly grew to innervate sweat glands in the skin. These fibers originally traveled via the:

a. Chorda tympani nerve
b. Greater petrosal nerve
c. Auriculotemporal nerve (derived from CN IX / otic ganglion pathway)
d. Long buccal nerve
e. Greater occipital nerve

22. A patient presents with an intraoral ulcer on the lateral margin of the tongue that is highly painful. The nerve pathways carrying this severe pain sensation from the anterior two-thirds of the tongue body travel via the:

a. Lingual nerve (branch of V3)
b. Chorda tympani nerve
c. Glossopharyngeal nerve
d. Hypoglossal nerve
e. Internal laryngeal nerve

23. If a patient presents with a localized lesion of the facial nerve *within* the petrous temporal bone *distal* to the stapedius branching but *proximal* to the exit of the chorda tympani, the clinical signs will manifest as:

a. Normal facial expressions but total loss of tearing
b. Loss of taste on the anterior two-thirds of the tongue and reduced salivation, alongside complete facial muscle paralysis
c. Pure hyperacusis with zero muscle weakness
d. Loss of general pain sensation on the tongue base
e. Deviation of the uvula toward the side of the lesion

24. During a deep dissection of the carotid triangle to clear a neoplastic mass, a surgeon accidentally transects the superior root of the ansa cervicalis (descendens hypoglossi). This will result in a loss of motor tone and subsequent paralysis of which group of neck muscles?

a. Suprahyoid muscles
b. Muscles of mastication
c. Infrahyoid strap muscles (except thyrohyoid)
d. Extrinsic tongue muscles
e. Intrinsic laryngeal muscles

22. A neurosurgeon performs a close evaluation of a patient with an upper motor neuron (UMN) lesion of the hypoglossal nerve (such as from a cortical stroke). When asked to protrude their tongue, the tongue tip deviates:

a. Toward the ipsilateral side of the lesion
b. Toward the contralateral side of the lesion (opposite the cortical stroke site)
c. Straight up vertically without deviation
d. It undergoes immediate flaccid wasting and fasciculations
e. It cannot be protruded past the premolars at all

26. A patient experiences an aneurysm of the internal carotid artery within the cavernous sinus. This vascular expansion selectively compresses a cranial nerve that runs directly through the center of the sinus alongside the artery. The patient will present with an inability to:

a. Close the eye tightly
b. Open the mouth against resistance
c. Abduct the eyeball laterally (due to CN VI compression)
d. Sense touch on the forehead skin
e. Protrude the tongue straight

27. A patient presents with a severe intracranial infection that tracks through the jugular foramen, creating a combined compression of cranial nerves IX, X, and XI (Vernet's syndrome). The patient will manifest which clinical triad?

a. Trismus, facial palsy, and blindness
b. Loss of taste on the posterior tongue, dysphagia (difficulty swallowing) with palatal drop, and inability to shrug the shoulder
c. Tongue deviation, loss of smell, and complete ptosis
d. Hyperacusis, loss of tearing, and anesthesia of the cheek mucosa
e. Miosis, partial ptosis, and loss of facial sweating

28. A patient presents with an emergency condition where a large fishbone is stuck deep within the piriform recess of the laryngopharynx. During attempts to retrieve the bone extraorally, a clinician must be careful to avoid injuring a nerve that runs immediately beneath the thin mucosa of the recess, which is the:

a. External laryngeal nerve
b. Recurrent laryngeal nerve
c. Internal laryngeal nerve (branch of CN X)

d. Glossopharyngeal nerve trunk
e. Hypoglossal nerve root

29. Direct injury to the internal laryngeal nerve within the piriform recess will result in a complete loss of which vital protective reflex mechanism?

a. Pupillary light reflex
b. Corneal blink reflex
c. Jaw-jerk reflex
d. Cough reflex (due to loss of sensory awareness above the vocal folds)
e. Salivary secretion reflex

30. A patient presents with a rare condition called glossopharyngeal neuralgia. Similar to trigeminal neuralgia, it features paroxysms of agonizing pain, but the triggers are swallowing or talking, and the pain maps to the tonsillar fossa and deep ear. To block this pain path, a surgeon must target fibers entering the:

a. Trigeminal ganglion
b. Jugular foramen (CN IX sensory pathway)
c. Foramen ovale
d. Foramen rotundum
e. Stylomastoid foramen

31. A patient presents with complete anesthesia over the skin of the lower lip and chin following a severe fracture through the stable body of the mandible. This clinical sign isolates direct traumatic compression or tearing of the:

a. Long buccal nerve
b. Lingual nerve
c. Mental nerve (terminal branch of the inferior alveolar nerve)
d. Mylohyoid nerve
e. Cervical branch of CN VII

32. A 62-year-old patient undergoes a radical carotid endarterectomy. Postoperatively, the patient experiences difficulty swallowing and a noticeable drop in blood pressure stability. The surgeon suspects iatrogenic damage to the pharyngeal branches and carotid sinus branches of the:

a. Trigeminal nerve
b. Facial nerve
c. Glossopharyngeal and vagus nerves (CN IX and CN X)
d. Hypoglossal nerve
e. Spinal accessory nerve

33. When testing the corneal reflex, a cotton wisp is touched to the transparent cornea. The patient immediately blinks both eyes. If a patient has a lesion of the left facial nerve motor root, what will happen when the left cornea is touched?

a. The left eye blinks but the right eye remains still
b. Neither eye blinks because the afferent path is blocked
c. The right eye blinks consensual style, but the left eye does not blink due to efferent motor loss
d. The patient experiences severe electric shock pain along V1
e. The pupil of the left eye dilates instantly

34. A patient presents with a severe neuro-neoplastic mass filling the pterygopalatine fossa. This lesion compromises the pterygopalatine ganglion, leading to a complete shutdown of secretomotor function affecting the:

a. Parotid gland

b. Submandibular gland
c. Lacrimal gland and minor mucosal glands of the nose and palate
d. Sublingual gland
e. Glands of Von Ebner

35. A patient experiences an isolated lesion of the chorda tympani nerve before it joins the lingual nerve in the infratemporal fossa. Quantitative salivary testing will reveal a marked reduction in daily saliva volume because this nerve controls secretion from the:

a. Parotid gland exclusively
b. Submandibular and sublingual salivary glands
c. Labial minor glands only
d. Lacrimal gland matrix
e. Glands of Bowman

36. A professional opera singer presents with an inability to produce high-pitched, tense vocal notes following a surgical resection of the superior pole of the thyroid gland. Laryngoscopy shows a lack of tension in the vocal folds due to selective injury to the:

a. Internal laryngeal nerve
b. External laryngeal nerve (supplying the cricothyroid muscle)
c. Recurrent laryngeal nerve
d. Main vagal trunk
e. Glossopharyngeal motor loop

37. A 7-year-old child presents with a severe middle ear infection (otitis media) that has spread to cause a localized inflammation of the chorda tympani nerve as it runs across the tympanic membrane. The child will manifest:

a. Loss of general somatic pain sensation on the tongue tip
b. Loss of taste over the anterior two-thirds of the tongue on that side
c. Complete paralysis of the masseter muscle
d. Ptosis of the upper eyelid
e. Deafness in both ears

38. If the spinal accessory nerve (CN XI) is completely severed inside the posterior triangle of the neck during a lymph node biopsy, the patient will experience a complete inability to:

a. Rotate the head to the opposite side
b. Shrug the shoulder or abduct the arm past the horizontal plane due to trapezius paralysis
c. Protrude the tongue straight
d. Depress the mandible to open the mouth
e. Swallow solid food chunks

39. A patient presents with a deviated uvula that points sharply to the right side when they say "Ah". This clinical presentation confirms a lower motor neuron lesion affecting the:

a. Right vagus nerve
b. Left vagus nerve (CN X supplying the left levator veli palatini muscle)
c. Right glossopharyngeal nerve
d. Left glossopharyngeal nerve
e. Right hypoglossal nerve

40. A clinician performs a local anesthetic block of the mandibular nerve (V3) at the foramen ovale. This successfully blocks sensory awareness from the lower jaw teeth and skin, but it also induces temporary paralysis in which group of muscles?

a. Muscles of facial expression
b. Infrahyoid strap muscles
c. Muscles of mastication (masseter, temporalis, pterygoids) along with tensor tympani, tensor veli palatini, mylohyoid, and anterior digastric
d. Intrinsic and extrinsic tongue muscles
e. Pharyngeal constrictors